

KAVINILAVU.S

CSA 1606 – Data

Warehousing + Dexta Mining

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CO-1-AT-I

1. Data Warehouse

Three popular

1. Star schema :

: Time, Doctor, Patient.

schemas used in Data Warehouses.

- One central fact table

connected to multiple
dimension tables.

->

→ Simple + fast for Querying

2. Snow flake Schema :

- Extension of Star Schema
-) Dimension tables are normalized into multiple related tables

3. Fact Constellation (galaxy schema):

=)

- Multiple fact tables share dimension tables.
-) Used for complex data Warehouses

(b) Star Schema Diagram

Time Time-ID (PK)

Day

Month

Year

Doctor

-1 . . . Fact Fee

Fee-.

Patient

Time. ID (FK)

Doctor - ID (FK)

Patient ID (FK)

Count

Charge

Doctor Table

Doctal- ID (PK)

Doctor Name

Hospital Specialization

Patient table

Patient. ID (PK)

Patient Name

Gender

Age

City

Fact Table: Fact-fee

Measures: Count, charge.

Dimensions: Time, Doctor,
Patient.

(c) SQL Query,

```
SELECT doctor,
```

```
year,
```

```
SUM ( charge) AS
```

```
total-charge,
```

```
SUM (count) AS Total - Visits
```

```
FROM fee
```

```
GROUP BY doctor year;
```

8) Big University Data Warehouse :

๑)

snowflake Schema:

University

Department

Course

Student Fact grades

Instructor

Semester

Year

Student

+) Student ID

Dimension Tables:

Course

→ Course ID

-> Name

→ Course – Name

Major

7

Fact Table :

Fact- epades

Student - ID

Course-ID

Semester - ID

Instructor – ID

4

– Status

Semester

Count

→ Semester_ID

Avg. grade

→

Semester

-

→ Year

T

→ Department

Instructor

→ Instructor_ID

T

Name

Department.

b) Required OLAP
Operations :

Goal:

Average grade of cs
courses for each student
Operations:

1. Slice:

D

-

Select Department - "CS".

2. Roll-up

Semester - year (optional)

3. Dice

1

delect only Required

students if needed.

Aggregate :

|

Compute AVG Cegrade).

mentries

حبا

Result :

Student

Avg Grade

Ammit

84

John

91

Riya

84.

(c) No. of Cuboids:

Formula:

(No. of levels) ^

Dimensions

fact Dimension: 5 levels

Dimensions

4

5

= 605 cuboids.

Data Warehouse Design for Regional Weather Bureau.

Star Schema :

Fact Table

Weather Fact :

Probe – ID

Location ID

Time ID

Pressure Temperature Precipitation.

Measures:

→ Temperature

→ Air Pressure

>

Precipitation

Dimension Tables,

Location :

Time :

Location ID

Time . ID

Hour

City

Region

Day

Probe: 04 (1

Probe - ID

Probe - Type

Installation - Date

Country

Status

Month Year

Land /Ocean

Star Acheng :

Benefits :

Time

1

Location - . . Fact Weather ---
Probe .

→ Fast OLAP Analysis

→ Trend Analysis over 10 years

→

Weather Forecasting

- +

→ Monthly & yearly summaries

4 . Data Cube :

Given:

t
 T_T
 T

n dimensions

→P distinct

a) Max

No concept

values per dimension hierarchy

no. of cells in Base Cuboid.

each dimension has p values.

Max cells = p^n

b) Min.no.

of cells in base cuboid:

If only

one valid tuple exists,

Min cells =
=1

c) Max no. of cells in

each dimension

Entire Data Cube :

has $p+1$.

Max cells $(p+1)^h$

d) Min

no.

=

of cells in Entire Data Cube :

If there is only one
base tuple, each cuboid
contains exactly one

Δ

Min cells = 2^n .
aggregate cell